Yijie He

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Education

University of California San Diego — San Diego, CA
M.S. in Computer Engineering — *Sep 2025 – March 2027 (Expected)*
B.S. in Computer Engineering — *Sep 2021 – Jun 2025*

Technical Skills

Languages: C/C++, Python, Java, SystemVerilog, OpenCL
Frameworks & Tools: PyTorch, CUDA, Triton, HIP/ROCm, vLLM, SGLang, Linux, Docker, Git

Work Experience

ByteDance — **Shanghai, China**
AI Inference System Intern, AML-Volcano Ark — *Jul 2026 - Present*

- Combined **VSA/Sol Attention** for Seedance 2.0/2.5 on AMD GPUs, mixing exact and summary-based computation with INT8 QK / FP8 PV HIP kernels. Optimized memory reuse, routing, and partitioning for **1.82× speedup over VSA** with no observed video quality degradation.
- Optimized **Conv1D** with shape-specific dispatch, fused layout/dtype conversion, prepacked weights, and stride-based transposed convolution decomposition. Exceeded MIOpen performance by ~**32%** on target workloads and eliminated solution-switching variability for **bitwise-deterministic outputs**.
- Optimized **Conv2D in HIP/CK** through tiling, vectorized layout conversion, weight prepacking, and direct output writes; exceeded MIOpen performance by ~**20%** on target workloads with **deterministic outputs**.
- Fused **All-to-All MoE communication and computation** with tiled asynchronous pipelines, L2 cache reuse, and triple buffering, achieving **1.84× speedup** over the original implementation in the target SP8 configuration.

Qualcomm — **San Diego, CA**
Software Engineering Intern — *Jun 2025 – Sep 2025*

- Ported and optimized **FlashAttention-2** for Qualcomm on-device GPUs through tiling, memory-access tuning, and local cache reuse. Eliminated register spills through low-level assembly optimization, improving throughput by **34%** on the same target workloads.
- Used internal profilers and **hardware instrumentation** to diagnose memory bottlenecks and guide kernel revisions; benchmarked specialized implementations on-device to verify performance gains.

Projects

AMD Kernel Design Agent — Aug 2026 – Present

- Built a **HIP/ROCm kernel optimization agent** with benchmark constraints on generation, validation, profiling, and review; combined compiler/runtime feedback with AMD hardware knowledge and tuning rules for iterative, cross-architecture optimization.
- Validated the workflow on **MoE**, **convolution**, and **VSA**; ported MoE to MI300X, exceeding the official implementation by **10%**, and achieved **1.08× speedup** over an optimized internal VSA kernel.

MiniMax H3 Inference & Framework Optimization | vLLM-Omni — Sep 2026 – Present

- Deployed H3 on AMD GPUs, fixed replica configuration fallback, and implemented **workload-aware routing** with load reservation, release, and rollback. Improved completed-request throughput by **11.44%**
- Implemented **merged QKV exchange and QKNorm/RoPE-to-send-buffer fusion**, inspired by Sol-H3, reducing communication calls and intermediate memory traffic. Achieved **3.19× speedup** over the original compute-and-pack path at a fixed shape.
- Adapted **in-flight cancellation** using shared-memory signals and coordinated rank exit; fixed audio decoding nondeterminism. Reduced time to complete all valid requests by **21%** in eight-GPU tests with **25% early cancellations**, verifying matching audio/video outputs.

LLM Core Kernel Replication and Deployment | Triton, CUDA — Feb 2026 – Mar 2026

- Reconstructed **Grouped GEMM and FlashAttention-2** in CUDA/Triton; resolved **NVFP4 GEMM compatibility on RTX 5090** using CUTLASS by adapting data layouts and compute paths.
- Implemented CUDA **FP4 encoding/decoding** and workspace preallocation to enable **Tensor Core/TMA acceleration**, achieving ~**2× speedup over BF16** on target workloads.